Sakartvelo Defenders

Game Design Bible v3.0

Chronologically Accurate Georgian History Tower Defense

10 Eras | 200 Levels | Blockchain-Powered

Godot 4.x | PWA | Immutable zkEVM

Confidential Design Document

Table of Contents

1. Vision **1**

2. Game Overview **2**

2.1 Core Gameplay Loop 2

2.2 Genre and Platform 2

2.3 Target Audience 3

3. Eras **3**

3.1 Era 0: Ancient Colchis (~1500 BC – ~300 BC) 3

3.2 Era 1: Kingdom of Iberia (~300 BC – ~630 AD) 4

3.3 Era 2: Age of Invasions (~630 – ~1089 AD) 4

3.4 Era 3: The Georgian Golden Age (~1089 – ~1225 AD) 5

3.5 Era 4: The Mongol Catastrophe (~1225 – ~1500 AD) 5

3.6 Era 5: Between Empires (~1500 – ~1801 AD) 6

3.7 Era 6: Russian Empire (~1801 – ~1918 AD) 6

3.8 Era 7: First Democratic Republic (1918 – 1921 AD) 7

3.9 Era 8: The Soviet Century (1921 – ~1991 AD) 7

3.10 Era 9: Modern Georgia (~1991 – Present) 8

4. Tower System **8**

4.1 Tower Upgrade System 9

4.2 Tower Synergy System 9

5. Enemies **10**

5.1 Enemy Categories 10

5.2 Boss Encounters 10

6. Heroes **11**

6.1 Hero Abilities and Cooldowns 11

6.2 Hero Progression 11

7. Maps **12**
7.1 Map Themes by Era 12
7.2 Elevation Mechanic 12
8. Tutorial **13**
9. Progression **13**
9.1 Era Unlock Requirements 14
9.2 Star System 14
10. Scrolls and Mastery **14**
10.1 Standard Scrolls 15
10.2 Mastery Scrolls (ERC-721) 15
10.3 SAKART Boost Scroll 16
11. Skins and Binding **16**
11.1 Skin Binding 16
11.2 Marketplace Tooltip 17
12. Player vs. Player (PvP) **17**
12.1 Attack-Then-Defend Format 17
12.2 Loadout System 17
12.3 Defense Setup 18
12.4 Hero and Scroll Selection 18
13. Survival Mode **18**
14. Seasons **19**
15. Endgame **19**
16. Educational System **20**
16.1 Cultural Footer 20
16.2 Historical Accuracy Policy 21
17. Mythology Integration **21**
18. SAKART Token Economy **21**
18.1 Token Distribution 22
18.2 Burn Mechanisms 22

18.3 Anti-Inflation Design....22
19. Fees and Revenue Model....23
19.1 Treasury and Developer Withdrawal....23
20. Smart Contracts....23
20.1 SAKART Token Contract (ERC-20)....24
20.2 Skin Contract (ERC-1155)....24
20.3 Mastery Scroll Contract (ERC-721)....24
21. Blockchain Integration....24
21.1 Wallet Integration....25
21.2 Chain Outage Fallback....25
21.3 Invisible Blockchain Philosophy....25
22. Multi-Account Prevention....25
23. Competitive Analysis....26
23.1 Web2 Tower Defense Landscape....26
23.2 Web3 Gaming Landscape....27
24. Art Pipeline....27
24.1 Style Bible System....27
24.2 AI Generation Workflow....27
25. Tech Stack....28
25.1 Godot 4.x and PWA Architecture....28
25.2 Blockchain Infrastructure....28
26. Design Principles....29
26.1 No-Nerf Rule....29
26.2 Invisible Blockchain....29
26.3 Historical Accuracy....30
26.4 Solo Developer Feasibility....30
26.5 Accessibility....30
27. Roadmap....30
27.1 Phase 1: Foundation (Months 1-3)....31

27.2 Phase 2: Expansion (Months 4-6)....31
27.3 Phase 3: Blockchain Integration (Months 7-9)....31
27.4 Phase 4: Full Campaign (Months 10-14)....32
27.5 Phase 5: Polish and Phase 6: Launch....32
28. Changelog (v2.0 to v3.0)....32

1. Vision

Sakartvelo Defenders is a historically accurate tower defense game set across the full sweep of Georgian history, from the ancient Kingdom of Colchis to the modern Republic of Georgia. The game is built on three foundational pillars: education through gameplay, invisible blockchain integration, and cultural preservation. Players defend Georgian lands against historically authentic waves of invaders across ten distinct eras, each meticulously researched and visually distinct.

The core philosophy is that players should never feel like they are interacting with a blockchain product. All Web3 elements, including the SAKART token economy, NFT-based mastery scrolls, and tradable cosmetic skins, operate beneath the surface. The gameplay experience is indistinguishable from a premium Web2 tower defense title, with blockchain providing true asset ownership and player-driven economies as invisible infrastructure rather than visible friction.

Georgia's history is one of the most dramatic and underrepresented narratives in gaming. Positioned at the crossroads of Europe and Asia, Georgia has faced invasions from virtually every regional power: Persians, Romans, Arabs, Mongols, Ottomans, Persians again, and Russians. Each era produced unique military technologies, heroic figures, and cultural achievements that translate naturally into compelling tower defense mechanics. The game's ten-era structure ensures players experience this entire arc, learning Georgian history organically through hundreds of levels of strategic gameplay.

2. Game Overview

2.1 Core Gameplay Loop

The fundamental gameplay loop of Sakartvelo Defenders follows the classic tower defense formula, refined with historical authenticity and strategic depth. Players progress through levels by placing defensive towers along enemy approach paths, utilizing heroes with unique abilities, and activating consumable scrolls at critical moments. Each level presents a map inspired by real Georgian landscapes and fortifications, with enemy waves that reflect the actual historical invaders of each era.

The loop begins with level selection within an era. Players assess the map layout, choose their tower composition from their available roster, and place towers strategically during a build phase. Enemy waves then advance along predetermined paths, and players must manage their economy by earning resources from defeated enemies to build additional

towers or upgrade existing ones. Heroes provide active abilities that players trigger manually, adding a layer of real-time decision-making. Scrolls offer powerful one-time effects that can turn the tide of difficult engagements.

Completion is measured through a three-star system. Each star represents a progressively harder challenge tier within the same level: surviving all waves (1 star), completing with a life threshold (2 stars), and completing under a strict time or resource constraint (3 stars). Each star earns a one-time SAKART token reward, establishing a natural economy where progression drives token distribution without allowing replay farming.

2.2 Genre and Platform

Sakartvelo Defenders is a 2D isometric tower defense game. The isometric perspective provides clear visibility of enemy paths, tower ranges, and strategic positioning while maintaining the aesthetic warmth of illustrated historical environments. The 2D approach enables efficient asset production using AI-generated art with strict style bible consistency controls, making the project feasible for a solo developer.

The game is developed in Godot 4.x and deployed as a Progressive Web App (PWA). This architectural choice eliminates server hosting costs entirely, as the game runs client-side in any modern browser. The PWA wrapper enables offline gameplay for the full campaign, with blockchain transactions queued locally during chain outages and submitted automatically when connectivity resumes. This ensures the player experience is never interrupted by infrastructure issues.

2.3 Target Audience

The primary audience encompasses tower defense enthusiasts, Web3 gaming early adopters, and players interested in history and cultural education. Tower defense fans will appreciate the strategic depth of synergy combinations, hero ability management, and the challenge of three-star completion across hundreds of levels. Web3 players will find genuine utility in the SAKART token economy without the usual crypto-game friction of mandatory wallet connections or pay-to-win mechanics.

A secondary but equally important audience is the Georgian diaspora community and cultural preservation advocates. Georgia has a rich cultural heritage that is underrepresented in mainstream gaming. By presenting this history through an accessible, engaging format, the game serves as both entertainment and educational tool. Historical pop-ups appear between levels, offering context about real events, figures, and locations depicted in the gameplay.

3. Eras

The game spans ten chronological eras, each representing a major period in Georgian history. Every era contains 20 levels, totaling 200 levels across the full campaign. The era structure follows a strict chronological progression, ensuring historical accuracy and allowing players to experience the complete narrative arc of Georgian civilization. Each era introduces new tower types, enemy variants, hero characters, and synergy combinations, providing continuous mechanical evolution alongside historical progression.

The chronological approach is non-negotiable. Georgian history is a continuous story where events in one era directly shape the next. The Arab invasions of the 7th century created the conditions for David IV's reforms in the 11th century. The Mongol devastation of the 13th century led directly to the fragmentation that made Georgia vulnerable to Ottoman and Persian domination. By maintaining strict chronological order, the game preserves historical cause-and-effect relationships that would be lost with a thematic or arbitrary ordering.

Era	Name	Period	Levels	Key Figures
0	Ancient Colchis	~1500 BC – ~300 BC	20	Medea, Aeëtes, Jason
1	Kingdom of Iberia	~300 BC – ~630 AD	20	Pharnavaz, St. Nino, Vakhtang Gorgasali
2	Age of Invasions	~630 – ~1089 AD	20	Arab Emirs, Bagrationi Kings
3	The Georgian Golden Age	~1089 – ~1225 AD	20	David IV, Queen Tamar
4	The Mongol Catastrophe	~1225 – ~1500 AD	20	Mongol Generals, Fragmented Kings
5	Between Empires	~1500 – ~1801 AD	20	Erekle II, Persian/Ottoman Rivals
6	Russian Empire	~1801 – ~1918 AD	20	Annexed Nobility, Cultural Figures
7	First Democratic Republic	1918 – 1921 AD	20	Noe Jordania, Independence Fighters
8	The Soviet Century	1921 – ~1991 AD	20	Stalin (historical), Soviet Era Figures

Era	Name	Period	Levels	Key Figures
9	Modern Georgia	~1991 – Present	20	Zviad Gamsakhurdia, Saakashvili

3.1 Era 0: Ancient Colchis (~1500 BC – ~300 BC)

The opening era of Sakartvelo Defenders transports players to the ancient Kingdom of Colchis, the land of the Golden Fleece. This era draws from Greek mythological sources and archaeological evidence to recreate the earliest period of Georgian statehood. Colchis was a wealthy kingdom located on the eastern shore of the Black Sea, known for its gold mining, metalworking, and advanced agricultural practices. The Greek myth of Jason and the Argonauts reflects the historical reality of Greek interest in Colchian gold and trade routes.

Gameplay in this era introduces fundamental tower defense mechanics with a primitive, bronze-age aesthetic. Towers are stylized as wooden watchtowers, stone fortifications, and archer platforms. Enemy waves represent tribal raiders, early Greek expeditionary forces, and mythological creatures drawn from Georgian folklore such as the devi (giant demons). The environment features lush Colchian forests, river valleys, and the ancient city of Vani, one of the most important archaeological sites of the Colchian civilization.

The hero for this era is Medea, the Colchian princess from Greek mythology. In Georgian tradition, Medea is reclaimed as a powerful local figure rather than a Greek villain. Her abilities reflect her mythological association with magic and herbalism, providing healing buffs for nearby towers and poison damage to enemies. The educational pop-ups for this era cover the archaeological evidence for Colchian civilization, the reality behind the Golden Fleece legend (possibly a reference to gold-panning techniques using sheep fleeces), and the cultural exchange between Colchis and the ancient Greek world.

3.2 Era 1: Kingdom of Iberia (~300 BC – ~630 AD)

The Kingdom of Iberia (Kartli in Georgian) represents the formation of the Georgian state as a major regional power. This era spans over nine centuries, from the establishment of the Iberian kingdom under King Pharnavaz to the Arab conquest of Tbilisi in the 7th century. It encompasses the Roman-Parthian rivalry for control of the Caucasus, the Christianization of Georgia by Saint Nino in the 4th century, and the reign of King Vakhtang Gorgasali, who founded Tbilisi as the capital.

Towers in this era evolve from simple wooden structures to stone fortifications and early siege equipment. New enemy types include Roman legionaries, Parthian cavalry, Sassanid Persian forces, and early Arab raiders. The introduction of Christianity as the state religion

provides narrative context for the era, with churches and monasteries appearing as defendable structures on maps. Environment art shifts to reflect the growing urbanization of Iberia, with maps set in the ancient capital of Mtskheta, the fortress of Uplistsikhe, and the newly founded city of Tbilisi.

Three heroes are available in this era: King Pharnavaz (founder of the kingdom, provides kingdom-wide buff abilities), Saint Nino (healer and protector, with abilities that shield towers and slow enemies), and King Vakhtang Gorgasali (military commander with offensive abilities that boost tower damage in an area). The era concludes with the devastating Arab invasions that will set the stage for the next era.

3.3 Era 2: Age of Invasions (~630 – ~1089 AD)

The Age of Invasions covers nearly five centuries of continuous foreign pressure on Georgian lands. Following the Arab conquest of Tbilisi in 645 AD, Georgia entered one of the most challenging periods in its history. The Tbilisi Emirate became a center of Islamic power in the Caucasus, while Byzantine and later Seljuk Turkish forces competed for influence over the fragmented Georgian principalities. This era is defined by the slow, determined consolidation of power by the Bagrationi dynasty, who would eventually unify Georgia and launch its golden age.

Mechanically, this era introduces the concept of fortified positions and defensive sieges. Maps feature crumbling Iberian fortresses being held against numerically superior Arab and Seljuk forces. New tower types include fortified gatehouses and mounted archer platforms. Enemy waves are larger and more aggressive, reflecting the overwhelming military pressure of the period. The difficulty curve increases significantly, as players must defend against sustained multi-wave assaults from enemies with greater variety and strength.

The heroes of this era represent the resistance and resilience of the Georgian people during centuries of foreign domination. Bagrationi princes and kings who led the slow reconquest feature as playable heroes, alongside religious figures who maintained Georgian cultural identity during the occupation. The educational content covers the Tbilisi Emirate, the Battle of Shirimni, the Battle of Sasireti, and the broader geopolitical context of Arab-Byzantine-Seljuk rivalry in the Caucasus.

3.4 Era 3: The Georgian Golden Age (~1089 – ~1225 AD)

The Georgian Golden Age is the centerpiece of the game, representing the period when Georgia reached its greatest territorial extent, cultural flowering, and military power. Initiated by King David IV (David the Builder), who reformed the military, reunited the Georgian territories, and defeated the Seljuk Turks at the Battle of Didgori in 1121, this era

culminated under Queen Tamar (r. 1184-1213), during whose reign Georgia controlled territories from the Black Sea to the Caspian and from the Caucasus Mountains to northern Iran.

This era features the most diverse tower roster in the game. Advanced fortifications, including the massive stone castles that dot the Georgian countryside, are joined by specialized units reflecting the professionalized Georgian military of the period. Monaspa royal guard towers, elite cavalry platforms, and siege defense emplacements provide players with powerful defensive options. Enemy waves represent the various peoples who challenged Georgian hegemony: remnants of Seljuk forces, Turkmen tribes, and northern Caucasian mountaineers.

Both David IV and Queen Tamar are playable heroes in this era, as the user has confirmed that more heroes is better and both figures are essential to the period. David IV provides military leadership buffs that increase tower damage and reduce upgrade costs, reflecting his historical reputation as the great reformer. Queen Tamar offers diplomatic and governance abilities that generate bonus resources and provide temporary invulnerability to structures, reflecting her reign of prosperity. Additional heroes from this era may include the poet Shota Rustaveli and the military commander Zakaria Mkhargrdzeli.

3.5 Era 4: The Mongol Catastrophe (~1225 – ~1500 AD)

The Mongol Catastrophe marks the abrupt end of the Georgian Golden Age. Beginning with the first Mongol expeditions into the Caucasus in the 1220s and culminating in the complete subjugation of Georgia by 1243, this era represents the most devastating period in Georgian history. The Mongol conquest destroyed the centralized Georgian state, decimated the population, and ushered in nearly three centuries of fragmentation and decline. Georgia was divided into competing kingdoms and principalities that paid tribute to the Mongol Golden Horde and later the Ilkhanate.

Gameplay in this era shifts dramatically in tone. Players are no longer defending a unified, prosperous kingdom but rather struggling to survive overwhelming odds with limited resources. Tower placement is constrained by crumbling infrastructure, and enemy waves are massive Mongol armies with superior mobility and firepower. New mechanics introduced in this era include retreat and evacuation objectives, where players must protect civilian populations as they withdraw to mountain fortresses. The environment art reflects the devastation: burned cities, abandoned villages, and fortified mountain redoubts.

The hero for this era represents the last generation of unified Georgian resistance, figures who fought desperately against the Mongol advance before the kingdom fractured. Educational pop-ups cover the Mongol administrative system in Georgia, the economic exploitation that drained the kingdom's wealth, and the cultural resilience that kept Georgian

identity alive even under foreign domination. This era sets the stage for the fragmentation that defines the subsequent era.

3.6 Era 5: Between Empires (~1500 – ~1801 AD)

Between Empires covers the period of Georgian history when the fragmented Georgian kingdoms found themselves caught between the Ottoman Empire and Safavid (later Afsharid and Qajar) Persia. This era saw both devastating wars and remarkable cultural achievements. King Erekle II of Kartli-Kakheti stands as the towering figure of this period, attempting to unite the eastern Georgian kingdoms and navigate the impossible diplomatic situation between rival empires. Despite his efforts, including a disastrous alliance with Catherine the Great of Russia, Georgia was ultimately unable to maintain its independence.

Towers in this era reflect the evolving military technology of the early modern period. Gunpowder weapons appear for the first time, alongside traditional Georgian defensive structures. Maps are set in the three major Georgian kingdoms of the period: Kartli, Kakheti, and Imereti, each with its own architectural style and strategic challenges. Enemy waves alternate between Ottoman and Persian forces, each with distinct unit compositions and attack patterns that require different defensive strategies.

King Erekle II is the primary hero of this era, providing abilities that combine military leadership with diplomatic finesse. His special ability temporarily reduces all enemy damage, reflecting his historical skill at negotiation even in wartime. The educational content covers the Battle of Krtsanisi (1795), one of the most tragic events in Georgian history, where the Persian army of Agha Mohammad Khan devastated Tbilisi despite Erekle's desperate defense. This event directly led to Georgia's incorporation into the Russian Empire.

3.7 Era 6: Russian Empire (~1801 – ~1918 AD)

The Russian Empire era covers the period from Georgia's annexation by Russia in 1801 to the collapse of the empire in 1918. Despite the loss of political independence, this era was marked by a remarkable cultural renaissance. Georgian literature flourished with the works of Ilia Chavchavadze and Akaki Tsereteli, theater and music experienced a golden age, and Georgian national identity was consciously preserved and strengthened. The era also saw major infrastructure development, including the construction of the Transcaucasian Railway connecting the Black and Caspian Seas.

Gameplay in this era introduces industrial-age military technology. Towers include fortified machine gun positions, artillery emplacements, and barricaded urban streets. Enemy waves represent the various conflicts of the period: the Caucasian War against Imam Shamil's

forces, the 1905 Russian Revolution, and the chaos of World War I on the Caucasus Front. Maps are set in rapidly modernizing Tbilisi, the mountain villages of Svaneti, and the Black Sea port city of Batumi.

The heroes of this era are primarily cultural and intellectual figures rather than military commanders. Ilia Chavchavadze provides abilities that boost tower efficiency and resource generation, reflecting his role as a national leader and modernizer. Other potential heroes include the poet and publicist Akaki Tsereteli and the military leader Gia Abesalomashvili. The educational content covers the Georgian national awakening, the abolition of serfdom, and the growing independence movement that would culminate in the next era.

3.8 Era 7: First Democratic Republic (1918 – 1921 AD)

The First Democratic Republic of Georgia (1918-1921) was the first modern democratic state in the Caucasus and one of the first democratic republics to emerge from the ruins of the Russian Empire. Led by the Social Democratic (Menshevik) government of Noe Jordania, Georgia declared independence on May 26, 1918, and spent three chaotic years building a multi-ethnic, multi-party democratic state while fighting wars against Armenia, Azerbaijan, White Russian forces, and the Bolshevik Red Army.

Despite its brief existence, the First Republic achieved remarkable accomplishments: a constitution guaranteeing universal suffrage and minority rights, land reform, judicial independence, and recognition by major international powers including the League of Nations. The era ended when the Bolshevik Red Army invaded in February 1921, forcing the democratic government into exile. This brief democratic experiment remains a powerful symbol in modern Georgian politics and national identity.

Gameplay in this era features early 20th-century military technology: rifles, machine guns, and artillery. Enemy waves represent the multiple threats faced by the republic simultaneously. Maps are set in Tbilisi, the Black Sea coast, and the border regions where the republic fought its defensive wars. The hero is Noe Jordania, whose abilities focus on diplomatic coordination that temporarily strengthens all towers and generates emergency resources. Educational pop-ups cover the constitution, the international recognition campaign, and the tragic circumstances of the Bolshevik invasion.

3.9 Era 8: The Soviet Century (1921 – ~1991 AD)

The Soviet Century covers the seventy years of Georgian history within the Soviet Union. This era is presented with historical honesty, neither glorifying the Soviet period nor erasing its complexities. Georgia experienced both severe repression and significant modernization during this time. The most controversial figure is Joseph Stalin (born Ioseb Jughashvili in

Gori, Georgia), who rose from Georgian origins to become the Soviet dictator responsible for millions of deaths. The game handles Stalin by presenting historical facts without commentary, allowing players to engage with this difficult history through the gameplay context.

Towers in this era reflect Soviet military technology: reinforced concrete bunkers, anti-aircraft positions, and fortified checkpoints. Enemy waves represent the various conflicts that touched Georgia during the Soviet period, including World War II (where approximately 350,000 Georgian soldiers died), anti-Soviet partisan resistance in the mountains, and the environmental and social disruptions of late Soviet industrialization. Maps are set in Soviet-era Tbilisi, industrial zones, and mountain villages that served as centers of resistance.

The hero selection for this era requires careful handling. Historical figures from the Soviet period are presented factually, with educational pop-ups providing context about their roles and the human cost of their actions. The game does not make Stalin a hero character in the playable sense but may reference him in historical context. Other potential heroes include World War II heroes, dissident writers, and cultural figures who preserved Georgian identity during the Soviet period. The educational content covers the Great Purge (which disproportionately affected Georgia), World War II, the 1978 language protests, and the growing independence movement of the late 1980s.

3.10 Era 9: Modern Georgia (~1991 – Present)

The final era brings players to the modern Republic of Georgia, from the chaotic first years of independence through the Rose Revolution of 2003 to the present day. This era covers the 1991-1993 civil war and war in Abkhazia, the economic collapse of the 1990s, the peaceful Rose Revolution that ousted the Shevardnadze government, the 2008 war with Russia, and the ongoing process of democratic consolidation and Euro-Atlantic integration.

Towers in this era represent modern military technology: precision-guided weapon systems, drone surveillance platforms, and fortified military outposts. Enemy waves reflect the conflicts of modern Georgia: separatist forces, irregular militias, and conventional military threats. Maps are set in modern Tbilisi, the conflict zones of South Ossetia and Abkhazia, and the strategic Georgian Military Highway connecting Tbilisi to Russia through the Caucasus Mountains.

This era's heroes are drawn from modern Georgian history, figures who shaped the country's transition from Soviet republic to independent democracy. The educational content covers the challenges of post-Soviet state-building, the achievements and controversies of the Saakashvili era, and Georgia's current strategic position as a Western-oriented democracy in a region of renewed Russian assertiveness. The final levels of this era serve as the game's

climax, bringing together all the mechanical systems and historical themes developed across the previous nine eras.

4. Tower System

The tower system is the mechanical core of Sakartvelo Defenders. Ten distinct tower types provide the player's defensive toolkit, each filling a specific strategic role. Towers are introduced gradually across the ten eras, with each era introducing one new tower type that reflects the military technology of that historical period. Every tower receives era-specific visual skins to maintain historical authenticity while preserving the underlying game mechanics that players have mastered.

The tower progression follows a clear design principle: newer towers add strategic options rather than rendering older ones obsolete. An Archer Tower from Era 0 remains viable in Era 9 because it occupies a unique niche (fast single-target damage at low cost) that no later tower perfectly replicates. This ensures that all ten tower types remain relevant throughout the full 200-level campaign, rewarding players who have mastered diverse tower compositions.

Tower Type	Role	Introduced	Era Skin Example
Archer Tower	Basic single-target DPS	Era 0	Colchian Watchtower / Iberian Fort
Catapult/Trebuchet	Area-of-effect splash damage	Era 0	Stone Thrower / Siege Engine
Wall/Barrier	Path blocking, enemy redirecting	Era 0	Wooden Palisade / Stone Wall
Cavalry Outpost	Fast patrol, intercepting enemies	Era 1	Iberian Horsemen / Georgian Knights
Siege Tower	Heavy damage, slow attack, anti-structure	Era 2	Fortified Gatehouse / Battlement
Magic/Shrine Tower	Buff/debuff, crowd control	Era 3	Church Shrine / Monastery
Gunpowder Tower	Long-range, piercing projectiles	Era 5	Cannon Emplacement / Musketry
Industrial Tower	Rapid-fire, multi-target suppression	Era 6	Machine Gun Nest / Artillery

Tower Type	Role	Introduced	Era Skin Example
Bunker Tower	Heavy armor, high survivability	Era 8	Soviet Bunker / Concrete Pillbox
Tech Tower	Precision strikes, drone support	Era 9	Modern Outpost / Drone Platform

4.1 Tower Upgrade System

Each tower can be upgraded through three levels, with each upgrade increasing damage, range, attack speed, or adding special effects. Upgrade costs scale exponentially, encouraging players to invest in a focused defense rather than spreading resources thinly across many weak towers. At maximum upgrade level (Level 3), towers gain a unique special ability specific to their type: Archer Towers gain a chance for critical hits, Catapults gain increased splash radius, and so on.

Tower upgrades are purchased with in-game resources earned from defeating enemies. This resource economy is entirely separate from the SAKART token economy and functions as the core gameplay loop driver. Players must make strategic decisions about when to invest in upgrades versus placing additional towers, creating meaningful choices every level. The upgrade system is designed to feel impactful: a Level 3 Archer Tower should feel dramatically more powerful than a Level 1, both visually and mechanically.

4.2 Tower Synergy System

Synergies are activated when specific tower combinations are placed within a 3-tile adjacency radius of each other. Synergies provide powerful bonuses that encourage diverse tower compositions and strategic map planning. Each synergy combination is introduced in the era where its component towers are most thematically appropriate, with new combinations becoming available as players unlock more tower types.

The synergy system creates emergent strategic depth. A player who places an Archer Tower near a Cavalry Outpost might discover a synergy that increases both units' attack speed, creating a rapid-response kill zone. Another player might combine a Magic/Shrine Tower with a Wall/Barrier to create a slowing zone that funnels enemies into a devastating Catapult kill zone. The system rewards experimentation and map-specific planning, ensuring that no single strategy dominates across all levels.

A comprehensive list of synergy combinations is documented in the Synergy Reference appendix (see Section 24). With ten tower types, there are 45 possible two-tower pairs and 120 possible three-tower triples, though only a curated subset of the most strategically interesting and thematically appropriate combinations will be implemented. Each era

introduces approximately 2-3 new synergy combinations, building up to a rich strategic toolkit by the final era.

5. Enemies

Enemy design in Sakartvelo Defenders follows two core principles: historical authenticity and mechanical clarity. Every enemy type is based on a real historical military unit or cultural archetype from Georgian history, modified as needed for gameplay balance. Enemies are visually designed to be instantly distinguishable, with clear silhouettes, color coding by threat level, and animation that communicates their behavior (fast enemies move erratically, armored enemies have deliberate, heavy animations).

Enemy HP and damage scale progressively across the 200 levels, but the scaling is carefully calibrated to maintain challenge without requiring numerical inflation that makes earlier towers irrelevant. Instead of simply increasing HP values, later eras introduce enemy types with new mechanics: shields that block frontal damage, healing abilities that restore HP to nearby enemies, flying units that bypass ground-based tower ranges, and boss enemies with unique phase-based attack patterns.

5.1 Enemy Categories

Enemies are divided into five broad categories that define their mechanical behavior. Infantry units follow standard path movement at moderate speed and HP, forming the backbone of most waves. Cavalry units move faster and have higher HP but follow the same paths as infantry. Siege units are slow but devastating, capable of damaging player towers directly. Flying units ignore path constraints and can only be targeted by towers with anti-air capabilities. Boss enemies appear at the end of specific levels as challenging multi-phase encounters.

Each era introduces era-specific variants within these categories. Era 0 enemies are tribal warriors and mythological creatures with low HP but fast movement. Era 2 enemies include heavily armored Arab cavalry and siege engines. Era 4 Mongol enemies feature devastating horse archers with hit-and-run tactics. Era 9 enemies include mechanized infantry and drone swarms that require modern tower capabilities to counter. The visual evolution of enemies across eras serves as both a gameplay progression mechanic and a visual history lesson.

5.2 Boss Encounters

Boss encounters appear at the end of every 5th level within each era (levels 5, 10, 15, and 20). Each boss is a named historical or mythological figure with a multi-phase fight that requires specific strategies to defeat. Bosses have dramatically higher HP than standard enemies and

feature unique abilities that challenge the player's tower placement and resource management.

Examples of planned bosses include the Colchian Dragon (Era 0), a mythical beast requiring coordinated multi-tower attacks; the Arab Emir of Tbilisi (Era 2), who summons reinforcing waves; the Mongol General Subutai's proxy (Era 4), who speeds up all enemy movement across the map; and Agha Mohammad Khan (Era 5), who deals area damage to nearby towers. Each boss fight is designed to be a memorable set-piece moment that rewards strategic thinking and diverse tower compositions.

6. Heroes

Heroes are special units that players deploy alongside their tower defenses. Each hero has a unique set of active abilities that the player triggers manually during combat, adding a layer of real-time decision-making to the tower defense formula. Heroes are era-specific, meaning they can only be used in levels belonging to their historical period. This design encourages players to engage with each era's historical context and ensures that hero abilities feel thematically appropriate.

The user has confirmed that more heroes is better. The design philosophy is to provide at least one hero per era, with major eras featuring two to three heroes to reflect the abundance of notable historical figures. As confirmed, both David IV and Queen Tamar are playable heroes in Era 3, representing the two most important figures of the Georgian Golden Age. Stalin is handled historically in Era 8, presented factually in educational content but not as a playable hero character.

Era	Hero	Type	Key Ability
0	Medea	Support/Control	Poison damage + tower healing
1	King Pharnavaz	Kingdom Buff	Economy boost, reduced tower costs
1	Saint Nino	Healer	Tower shield, enemy slow aura
1	Vakhtang Gorgasali	Offense	Area damage boost for nearby towers
3	David IV (David the Builder)	Offense/Leadership	Global damage buff, upgrade cost reduction

Era	Hero	Type	Key Ability
3	Queen Tamar	Defense/Economy	Resource generation, temporary tower invulnerability
3	Shota Rustaveli	Morale Boost	Attack speed increase for all towers
5	King Erekle II	Defense/Diplomacy	Enemy damage reduction, emergency resources
6	Ilia Chavchavadze	Economy/Culture	Tower efficiency, resource generation
7	Noe Jordania	Diplomacy	Temporary global tower strength boost
8	WWII Georgian Hero	Offense	Targeted strike ability
9	Modern Era Figure	Tech/Precision	Drone recon, precision targeting

6.1 Hero Abilities and Cooldowns

Each hero has three abilities: one basic ability on a short cooldown (15-20 seconds), one advanced ability on a medium cooldown (45-60 seconds), and one ultimate ability on a long cooldown (120-180 seconds). Basic abilities provide tactical benefits like minor damage buffs or healing. Advanced abilities can shift the flow of an individual wave. Ultimate abilities are game-changing effects that, when timed correctly, can save a failing defense or enable a perfect three-star run.

When a hero is defeated (all HP depleted), they enter a cooldown period before they can be redeployed. During this cooldown, the player must rely solely on their tower defenses. The cooldown duration scales with the hero's power level, ensuring that losing a powerful hero like Queen Tamar is more punishing than losing a support-oriented hero. This mechanic encourages players to position heroes strategically rather than exposing them to unnecessary damage.

6.2 Hero Progression

Heroes level up through gameplay by participating in battles. Each battle a hero participates in grants experience points based on the enemies defeated. Leveling up improves hero base stats (HP, damage) and reduces ability cooldowns. Hero levels are stored on-chain as part of the player's account data, ensuring that progression is persistent and portable. The maximum

hero level is capped to prevent power inflation, with the cap set to ensure that a max-level hero is powerful but not capable of solo-carrying a level without tower support.

7. Maps

Each of the 200 levels is set on a unique map inspired by real Georgian landscapes, fortifications, and cities. Maps are designed with the isometric 2D perspective in mind, ensuring clear visibility of enemy paths, tower placement zones, and strategic elevation points. The art style for each map reflects the architectural and environmental characteristics of the era, creating a visual journey through Georgian history that parallels the mechanical progression.

Map design follows several key principles. First, every map must have at least three viable tower placement strategies to encourage replay and experimentation. Second, maps must feature at least one elevation mechanic, where towers placed on higher ground receive increased range, adding a spatial dimension to tower placement decisions. Third, each map must include at least one chokepoint where players can concentrate defensive fire, balanced by at least one wide-open section that requires spread defenses. These principles ensure strategic variety across all 200 levels.

7.1 Map Themes by Era

Era 0 maps feature prehistoric Colchian landscapes: dense forests, river crossings, and the ancient gold-panning sites of the Rioni River. Era 1 maps showcase the early urban centers of Mtskheta and Uplistsikhe, with cave-city layouts providing unique pathing challenges. Era 2 maps depict fortified monasteries and mountain passes under siege from Arab and Seljuk forces. Era 3 maps feature the grand castles and cathedrals of the golden age, including Gelati Monastery and the cave city of Vardzia.

Era 4 maps show the devastation of the Mongol period: burning cities, abandoned fortresses, and desperate mountain retreats. Era 5 maps are set in the three kingdoms of Kartli, Kakheti, and Imereti, with distinct architectural styles for each. Era 6 maps feature rapidly modernizing Tbilisi with its mix of old and new architecture. Era 7 maps depict the war-torn landscapes of the independence period. Era 8 maps showcase Soviet-era industrial zones and mountain villages. Era 9 maps present modern Georgia with contemporary military installations and urban environments.

7.2 Elevation Mechanic

The elevation mechanic is a core map feature that rewards strategic tower placement. Maps are divided into ground-level and elevated positions, with higher positions providing a range

bonus to any tower placed there. The range bonus is approximately 15-20%, giving elevated towers a meaningful but not overwhelming advantage. Maps are designed so that the best elevated positions are also the most exposed or hardest to defend, creating risk-reward decisions for players.

Additionally, arrow-type and projectile towers benefit disproportionately from elevation, gaining both increased range and a damage bonus due to the improved angle of attack. This means an Archer Tower on a hill is dramatically more effective than one on flat ground, encouraging players to prioritize elevated positions for their projectile-based defenses while using melee-adjacent towers like Wall/Barrier units to control ground-level approaches.

8. Tutorial

The tutorial is integrated into the first five levels of Era 0, teaching core mechanics organically through gameplay rather than separate tutorial screens. The philosophy is that players learn by doing, with contextual hints appearing only when the game detects that a player is struggling. The tutorial levels are designed to be completable by any player regardless of experience with tower defense games, while still being engaging enough to retain experienced players.

Level 1 introduces tower placement: the player is given a single Archer Tower and must place it to defend against a slow, weak enemy wave. Level 2 introduces the economy system, requiring players to earn resources from kills and invest them in a second tower. Level 3 introduces tower upgrades by presenting a wave that cannot be defeated with Level 1 towers alone. Level 4 introduces the hero system with Medea's basic ability. Level 5 introduces scrolls as a consumable resource and features the first mini-boss encounter. By the end of these five levels, players have mastered all core systems and are ready for the full campaign experience.

Each tutorial level includes a brief historical pop-up between levels that introduces the world of Ancient Colchis. These pop-ups are concise (2-3 sentences) and focus on creating intrigue rather than delivering encyclopedic information. The deeper educational content is reserved for the main campaign, where players who are already invested in the gameplay will be more receptive to historical context.

9. Progression

Progression in Sakartvelo Defenders operates on two parallel tracks: gameplay progression and economic progression. Gameplay progression is the journey through 200 levels across 10 eras, with each level unlocked by completing the previous one. Economic progression is

the accumulation of SAKART tokens earned from star completions, which can be used to purchase cosmetic items, mastery scrolls, and access to competitive features.

The progression system is designed to be generous with gameplay content while strictly controlling economic rewards. All 200 levels are accessible to every player without any payment requirements. The SAKART economy is driven entirely by in-game achievement: each star earned on a level grants a one-time token reward, with higher stars granting larger rewards. Because each star can only be earned once per level, there is zero incentive for replay farming, and the total token supply is mathematically bounded by the number of players and levels.

9.1 Era Unlock Requirements

Eras are unlocked sequentially. To unlock the next era, a player must earn at least 1 star (the basic completion star) on a minimum number of levels in the current era. The requirement is set at 15 out of 20 levels, allowing players to skip the most difficult levels while still demonstrating sufficient mastery. This ensures that players experience most of each era's content before moving forward, maintaining the chronological educational narrative without creating frustrating progression blockers.

The era unlock requirement is intentionally permissive rather than restrictive. The game's philosophy is that players should be encouraged to explore, not punished for difficulty. A player who struggles with specific levels can move forward after completing the majority, then return later with better towers, higher-level heroes, and more experience to tackle the challenges they skipped. The three-star system provides the difficulty ceiling for dedicated players, while the one-star unlock provides the accessible floor for casual players.

9.2 Star System

Each level offers three stars, representing three tiers of completion challenge. The first star is earned by surviving all enemy waves without losing all lives. The second star requires completing the level while retaining a minimum percentage of lives (typically 50% or more). The third star requires completing the level under an additional constraint, which varies by level and may include time limits, resource caps, tower count limits, or no-upgrade restrictions.

Each star grants a one-time SAKART reward. The reward amounts scale with the difficulty of the era: early era stars grant fewer tokens, while later era stars grant progressively more. This scaling reflects both the increased difficulty of later content and the economic design goal of distributing tokens over the long term rather than front-loading rewards in early

content. Once a star is earned, replaying the level generates zero additional SAKART, eliminating replay farming as a concern entirely.

10. Scrolls and Mastery

Scrolls are consumable items that provide powerful temporary effects during gameplay. They are the primary tactical resource that players manage during a level, alongside tower placement and hero ability usage. Scrolls are single-use: activating a scroll consumes it permanently, and players must acquire new scrolls to replenish their inventory. Each player account can hold a maximum of 50 scrolls at any time, preventing hoarding and encouraging strategic use.

Scrolls are divided into two categories: standard scrolls and mastery scrolls. Standard scrolls provide gameplay effects during levels (damage boosts, resource generation, crowd control, healing, etc.) and are earned through gameplay progression or purchased with SAKART tokens. Mastery scrolls provide permanent passive bonuses to specific tower types or heroes and are represented as ERC-721 NFTs on the blockchain, making them tradeable on the marketplace.

10.1 Standard Scrolls

Standard scrolls come in several types, each providing a different tactical effect. Damage scrolls boost tower damage output for a limited duration. Economy scrolls generate bonus in-game resources. Crowd control scrolls freeze or slow all enemies on the map. Healing scrolls restore HP to damaged towers. Shield scrolls grant temporary invulnerability to a targeted tower or hero. Each scroll type has multiple tiers of potency, with higher tiers providing stronger effects.

Standard scrolls are consumed when activated during a level. They are not blockchain assets and cannot be traded between players. This design keeps the tactical resource economy simple and predictable. Standard scrolls can be earned as rewards for completing levels with 3 stars, purchased from an in-game shop using in-game currency (not SAKART), or occasionally received as daily login bonuses. The scroll inventory cap of 50 ensures that players must make meaningful choices about which scrolls to bring into each level.

10.2 Mastery Scrolls (ERC-721)

Mastery scrolls are NFT-based permanent passive bonuses. Unlike standard scrolls, mastery scrolls are not consumed during gameplay. Instead, they are activated in the player's loadout before a level begins and provide their bonus for the entire level duration. Mastery scrolls are type-based, not unique: there is no limit on how many copies of a specific mastery scroll can

exist, but each player can only activate each type of mastery scroll once per level, with a maximum stack of 3 copies of the same type providing increasing bonuses.

The stack system works as follows: activating 1 copy of a mastery scroll provides a 3% bonus to the relevant stat, 2 copies provide 6%, and 3 copies provide the maximum of 9%. This cap ensures that mastery scrolls enhance gameplay without breaking balance. For example, a player with 3 copies of the 'Archer Tower Mastery' scroll would have their Archer Towers deal 9% more damage for the duration of any level where the scroll is activated. Mastery scrolls are tradeable on the marketplace, creating a player-driven economy for these permanent upgrades.

10.3 SAKART Boost Scroll

As confirmed during design review, there is NO SAKART Boost scroll. The token economy is driven entirely by star-based earning, and no consumable item should exist that accelerates or modifies SAKART rewards. This prevents economic exploits and keeps the token distribution system clean and predictable.

11. Skins and Binding

Cosmetic skins are the primary visual customization system and the main use case for the SAKART token economy. Skins change the visual appearance of towers and heroes without affecting gameplay mechanics. They are implemented as ERC-1155 tokens on the Immutable zkEVM blockchain, enabling frictionless trading on the marketplace. Skins are purely cosmetic: a tower with a skin has identical stats and abilities to the same tower without a skin.

The skin system is designed around the era structure. Each tower type has one default skin per era (automatically applied when playing levels in that era) and multiple premium skins that players can acquire through gameplay rewards or marketplace purchases. Premium skins feature more detailed art, special effects, and animation variations that make them visually distinct from the default era skins.

11.1 Skin Binding

When a player equips a skin to a tower, the skin becomes bound to that specific tower instance. Bound skins cannot be traded on the marketplace until they are unbound, which costs a small amount of SAKART. This binding mechanism serves two purposes: it creates a SAKART token sink (burn mechanism) through the bind/unbind fee, and it prevents marketplace speculation on skins that are actively being used in gameplay.

The bind/unbind fee is set low enough to not discourage players from experimenting with different skins, but high enough to create meaningful economic friction against rapid marketplace trading. The exact fee amount is subject to balance testing and may vary based on the skin's rarity tier. Common skins have minimal bind/unbind fees, while rare and legendary skins have higher fees, reflecting their greater marketplace value.

11.2 Marketplace Tooltip

As confirmed during the design review with Hermes, every item on the marketplace displays a clear tooltip showing: what the item does, which era/level it applies to, its current bind status, and the cost to unbind if applicable. This transparency ensures that players can make informed purchasing decisions without needing to consult external resources. The tooltip system is especially important for mastery scrolls, where the stack bonus progression (3%, 6%, 9%) must be clearly communicated.

12. Player vs. Player (PvP)

PvP in Sakartvelo Defenders is an asynchronous attack-then-defend format designed for fairness and accessibility. Unlike synchronous PvP tower defense, which requires both players to be online simultaneously, the asynchronous format allows players to attack an opponent's preset defense at their convenience, and defend against incoming attacks with AI-managed towers. This design respects the player's time while maintaining competitive depth.

12.1 Attack-Then-Defend Format

Each PvP engagement consists of two phases. In the Attack phase, the player selects a league-level map and sends enemy waves against the opponent's pre-built defense. In the Defend phase, the player's own defense is tested against waves sent by their opponent. The winner is determined by total score across both phases, where score is based on lives remaining, time survived, and enemies defeated.

The turn order for who attacks first is determined by a Rock-Paper-Scissors (RPS) mechanic at the start of each engagement. Both players secretly choose their throw, and the winner attacks first. Attacking first provides a slight psychological advantage (knowing the defense layout) but also reveals information that the defender can use in the second phase. This simple RPS mechanic adds a layer of pre-game strategy without introducing complexity that would alienate casual players.

12.2 Loadout System

PvP uses identical loadouts for both players to ensure absolute fairness. When a PvP engagement begins, both players are assigned the same tower roster, hero, and scroll selection. This eliminates pay-to-win concerns, as no player can gain an advantage through superior NFTs or token spending. The skill expression in PvP comes entirely from tower placement strategy, timing of hero abilities, and scroll usage.

There are no re-rolls on scrolls in PvP. As confirmed during design review, allowing re-rolls would create imbalance by letting players gamble for optimal scroll draws. Each player receives a fixed scroll selection at the start of the engagement and must make the best use of what they are given. This constraint encourages strategic planning and rewards players who can adapt their tactics to available resources rather than relying on lucky draws.

12.3 Defense Setup

Players build their defense on a standardized PvP map layout before entering matchmaking. The defense setup uses pre-built tower configurations that are executed by AI when the player is being attacked. The AI uses a predefined priority system: towers target the enemy closest to the exit, use abilities when enough enemies are in range, and do not make placement changes during the defense. This ensures that the defense performs consistently regardless of whether the player is online.

The PvP league system is organized by skill tiers rather than eras. As confirmed during design review, PvP should not have era restrictions; only league levels should gate matchmaking. This means a player who has completed Era 5 can compete in PvP against players at the same league level regardless of their campaign progress. The league system uses a standard ELO-based rating with promotion and demotion between tiers.

12.4 Hero and Scroll Selection

For the launch version, PvP uses a fixed rotating roster of heroes and scrolls that changes weekly. This design ensures balance without requiring continuous meta adjustments, and it keeps the PvP experience fresh by regularly changing the available strategic options. The rotating roster is selected by the development team based on balance testing, with community feedback incorporated through regular surveys. Future versions may introduce a draft-pick system where players alternately select heroes and scrolls from a shared pool.

13. Survival Mode

Survival Mode is an endless wave defense challenge separate from the main campaign. Players select a map and defend against increasingly difficult waves that continue until all lives are lost. There is no win condition in Survival Mode; the goal is to survive as many waves as possible and achieve the highest score on the global leaderboard. Survival Mode is the primary competitive format for players who have completed the campaign and seek ongoing challenges.

Survival Mode uses the player's full tower roster, hero collection, and scroll inventory from the campaign. Unlike PvP, there are no loadout restrictions, allowing players to use their best available assets. However, the scaling difficulty ensures that even the most powerful loadout will eventually be overwhelmed, typically around wave 50-100 depending on player skill. The global leaderboard tracks the highest wave reached, total enemies defeated, and score, with seasonal resets that award SAKART tokens to top performers.

14. Seasons

Seasons are quarterly content updates that introduce new gameplay features, limited-time events, and exclusive cosmetic rewards. Each season lasts approximately three months and is themed around a specific aspect of Georgian culture or history that may not fit into the main chronological campaign. For example, a season might focus on Georgian mythology (featuring folklore-based enemies and towers), Georgian cuisine (with food-themed cosmetic skins), or specific historical events like the Battle of Didgori.

Seasons serve several design purposes. They provide ongoing content for players who have completed the campaign, creating reasons to return regularly. They introduce variety into the gameplay experience by temporarily modifying core rules (such as double damage for a specific tower type, or altered enemy compositions). They generate marketplace activity through limited-edition cosmetic skins that are only available during the season. And they provide additional SAKART burn mechanisms through season pass purchases and exclusive item acquisition.

Each season includes a free track with basic rewards available to all players, and a premium track purchased with SAKART that offers enhanced rewards including exclusive skins, scroll packs, and mastery scroll fragments. The season pass price is set to create a meaningful but not punitive SAKART sink, contributing to the target annual deflation rate.

15. Endgame

The endgame begins when a player completes all 200 campaign levels and has access to the full tower roster, hero collection, and competitive modes. The endgame is designed to provide long-term engagement through three pillars: mastery (perfecting three-star completions and Survival Mode performance), collection (acquiring rare skins and mastery scrolls through the marketplace), and competition (climbing PvP and Survival leaderboards).

Mastery challenges include achieving three stars on all 200 levels, completing levels under self-imposed restrictions (no upgrades, single tower type only, hero-only runs), and setting high scores in Survival Mode. These challenges provide bragging rights and are tracked on the player's public profile. Collection goals revolve around acquiring the full set of cosmetic skins for each tower type and hero, including rare limited-edition skins from past seasons. Competition is maintained through seasonal leaderboard resets and regular PvP tournaments with SAKART prize pools.

The endgame economy is critical for long-term token health. With campaign star rewards largely earned by this point, the endgame introduces additional SAKART sinks through cosmetic purchases, season passes, and tournament entry fees. The balance between endgame sinks and remaining campaign earning ensures that the token economy remains healthy without requiring players to spend real money to continue enjoying the game.

16. Educational System

The educational system is one of the three foundational pillars of Sakartvelo Defenders. The game is designed to teach Georgian history organically through gameplay, with educational content woven into the fabric of the experience rather than presented as separate lessons. The educational system operates on three levels: passive learning through visual design and environmental storytelling, active learning through historical pop-ups between levels, and deep learning through an optional in-game encyclopedia.

Passive learning happens naturally as players progress through the campaign. The era-specific art styles, tower designs, enemy types, and map environments all communicate historical information visually. A player who spends 20 levels defending against Mongol armies in Era 4 will develop an intuitive understanding of the Mongol period's impact on Georgia, even if they never read a single educational pop-up. The visual design language is the most powerful educational tool in the game.

Historical pop-ups appear between every level and provide 2-4 sentences of context about the events, figures, or locations featured in the upcoming level. These pop-ups are skippable for players who want to focus on gameplay, but they are designed to be engaging enough that

most players will read them. As confirmed during design review, historical pop-ups do NOT cost SAKART to read. Making education paywalled would undermine the game's core philosophy of cultural preservation and accessibility.

16.1 Cultural Footer

As confirmed during the design review, the game includes a cultural footer on the main menu and loading screens that displays rotating Georgian cultural facts, proverbs, and historical quotes. This passive educational element adds flavor without interrupting gameplay and reinforces the game's identity as a cultural product. The cultural footer content is curated to be interesting and diverse, covering topics ranging from Georgian wine-making traditions (the world's oldest) to the unique Georgian alphabet, a UNESCO Intangible Cultural Heritage.

16.2 Historical Accuracy Policy

The game is committed to historical accuracy wherever possible. All historical events, figures, and cultural elements depicted in the game are based on accepted historical scholarship and Georgian historical sources. Where the historical record is ambiguous or contested, the game presents the most widely accepted interpretation and notes the existence of alternative views. The Stalin era (Era 8) is handled with particular care, presenting factual information about his actions and their consequences without glorification or erasure.

The historical accuracy policy extends to visual design. Tower skins, enemy designs, map environments, and hero appearances are all based on historical research and archaeological evidence. The Art Style Guide (see companion document) provides detailed reference material for each era's visual language, ensuring that the AI-generated art assets maintain consistency with historical reality. Anachronistic elements are avoided unless they serve a specific gameplay purpose and are clearly identified as non-historical.

17. Mythology Integration

Georgian mythology is deeply intertwined with the country's historical identity, and Sakartvelo Defenders incorporates mythological elements as complementary to the historical narrative rather than as a separate system. Mythological creatures and heroes appear primarily in Era 0 (Ancient Colchis, where the boundary between myth and history is naturally blurred) and in special seasonal events that explore specific aspects of Georgian folklore.

Key mythological elements include the devi (giant demons from Georgian folklore who serve as boss enemies), the kaji (magical blacksmiths who serve as tower upgrade mechanics

in the early eras), and the myth of the Golden Fleece (which provides the narrative framework for Era 0). The game also draws on the epic poem 'The Knight in the Panther's Skin' by Shota Rustaveli for heroic inspiration, with characters and themes from the poem influencing hero designs and ability concepts.

The mythology system is designed to enhance rather than replace the historical content. Players who are interested in mythology will find rich references and references throughout the game, while players who prefer pure history can focus on the historical eras without missing any core gameplay content. The distinction between mythological and historical content is always clearly communicated through the educational system, ensuring that players understand which elements are based on historical records and which are drawn from folklore.

18. SAKART Token Economy

SAKART is the ERC-20 utility token that powers the Sakartvelo Defenders economy. The token has a fixed hard cap of 1 billion (1,000,000,000) SAKART, with zero pre-mine and zero team allocation. All tokens enter circulation exclusively through player gameplay: stars earned on campaign levels are the sole distribution mechanism. This design ensures that the token economy is entirely player-driven, with no inflation from team unlocks, investor allocations, or algorithmic minting.

18.1 Token Distribution

Token distribution follows a star-based system. Each of the 200 campaign levels offers 3 stars (600 total stars). Each star, when earned for the first time, grants a one-time SAKART reward. The reward per star scales with the era, with early eras granting smaller rewards and later eras granting larger ones. The total token supply allocated to campaign rewards is approximately 700 million SAKART (70% of the hard cap), with the remaining 300 million reserved for seasonal rewards, competitive prizes, and marketplace liquidity incentives.

Because each star can only be earned once per player, the total number of SAKART tokens in circulation is directly proportional to the number of active players and their campaign progress. A player who earns all 600 stars across the full campaign will have accumulated a substantial but finite amount of SAKART. There is no mechanism for earning SAKART through replay, grinding, or any activity other than first-time star completions. This structural design makes replay farming mathematically impossible.

18.2 Burn Mechanisms

The SAKART economy includes multiple burn mechanisms designed to create sustained deflationary pressure. The primary burn mechanisms are: skin purchases on the marketplace (marketplace fees are partially burned), skin bind/unbind fees, season pass purchases, and cosmetic effect unlocks. Additional burn mechanisms under consideration include scroll reroll fees (allowing players to reroll their scroll selection for a level at a SAKART cost), cosmetic tower effects (visual effects applied to towers that cost SAKART per activation), and leaderboard entry fees for competitive tournaments.

The target annual deflation rate is 2-5% of circulating supply, achieved through the combination of all burn mechanisms. This deflation is not aggressive enough to create hoarding behavior (which would harm the gameplay economy) but sufficient to offset the slow trickle of new tokens entering circulation as new players join and earn stars. The fee structure is designed to be invisible to casual players while providing meaningful economic activity for engaged players.

18.3 Anti-Inflation Design

The SAKART economy is structurally anti-inflationary due to several design choices. First, the star-based earning system caps total possible earnings per player. Second, the absence of replay farming eliminates the most common Web3 game inflation vector. Third, the fixed hard cap of 1 billion tokens provides an absolute ceiling on supply. Fourth, the multiple burn mechanisms create continuous deflationary pressure. Fifth, the gameplay economy (in-game resources for tower placement and upgrades) is entirely separate from the SAKART economy, preventing token inflation from affecting gameplay balance.

19. Fees and Revenue Model

The revenue model for Sakartvelo Defenders is built around a transparent transaction fee applied to all marketplace trades. The total fee is 1.8% per transaction, distributed as follows: 1.3% to the development treasury for ongoing game development, server costs (minimal due to PWA architecture), and team compensation; 0.5% automatically compounded into the marketplace liquidity pool to ensure healthy trading conditions. No additional fees are charged for gameplay, account creation, or any other activity.

The 1.8% fee structure is designed to be low enough to encourage active marketplace trading while generating sufficient revenue to sustain development. As a solo developer with a $100/month budget, the minimal server costs (PWA has zero server hosting) mean that marketplace fees are almost entirely profit. The automatic LP compounding (0.5%) ensures

that the marketplace remains liquid even during low-activity periods, preventing the common Web3 problem of illiquid markets that discourage trading.

19.1 Treasury and Developer Withdrawal

The development treasury accumulates the 1.3% developer fee from all marketplace transactions. Withdrawals from the treasury are subject to a monthly cap to prevent sudden large withdrawals that could harm community trust. The monthly cap is set to cover development costs plus a reasonable developer salary, with any excess remaining in the treasury for future development needs. Treasury activity is fully transparent and auditable on-chain.

The withdrawal mechanism requires a timelock: withdrawal requests are queued for 48 hours before execution, during which they are publicly visible on the blockchain. This transparency mechanism ensures that the community can verify that treasury funds are being used for legitimate development purposes. As a solo developer, this level of transparency is both a business practice and a competitive advantage, building trust with the player community that the game will be maintained and improved over the long term.

20. Smart Contracts

Sakartvelo Defenders deploys on the Immutable zkEVM Layer 2 network, which provides zero gas fee transactions for players while maintaining the security guarantees of Ethereum. The smart contract architecture consists of three primary contracts: the SAKART token contract (ERC-20), the Skin contract (ERC-1155), and the Mastery Scroll contract (ERC-721). All contracts follow established security best practices and will be audited before mainnet deployment.

20.1 SAKART Token Contract (ERC-20)

The SAKART token contract implements the standard ERC-20 interface with custom modifications for the game's requirements. The total supply is fixed at 1 billion tokens with no minting function. Token transfers to and from the game's designated contracts (marketplace, season pass, bind/unbind) are gasless through Immutable's account abstraction. The contract includes a blacklist function for fraud prevention and multi-account detection, governed by a multisig wallet controlled by the developer.

20.2 Skin Contract (ERC-1155)

The ERC-1155 standard is used for cosmetic skins because it supports both fungible and non-fungible tokens within a single contract. Common skins are semi-fungible (multiple identical copies exist), while rare and legendary skins are non-fungible (unique or limited edition). The contract includes metadata URI pointers to decentralized storage (IPFS) for skin artwork and description data, ensuring that skin assets remain accessible even if the game's servers go offline.

20.3 Mastery Scroll Contract (ERC-721)

Mastery scrolls use the ERC-721 standard because each scroll is a distinct non-fungible asset, even when multiple scrolls of the same type exist. The contract tracks scroll type, stack count, and activation status. When a player activates a mastery scroll for a level, the contract verifies ownership and stack count before applying the bonus. The contract also manages the bind/unbind mechanism, requiring a SAKART payment for unbinding that is automatically burned.

21. Blockchain Integration

The blockchain integration follows the 'invisible blockchain' design philosophy: players should never feel like they are interacting with a blockchain product unless they choose to engage with the marketplace or token features. All core gameplay, including the full 200-level campaign, PvP, Survival Mode, and hero progression, functions identically whether or not the player has connected a wallet. Blockchain features enhance the experience for players who want true asset ownership and tradable items, but they are never required for gameplay.

21.1 Wallet Integration

Wallet connection is handled through Immutable's Passport wallet abstraction system. Passport provides a seamless onboarding experience that allows players to create a wallet using social login (email, Google, Apple) without needing to understand blockchain concepts. For players who already have Web3 wallets, Passport supports direct connection through MetaMask, WalletConnect, and other popular wallet providers. The wallet connection is optional and can be deferred indefinitely without affecting gameplay access.

21.2 Chain Outage Fallback

As confirmed during design review with Hermes, the game implements a chain outage fallback system. If the Immutable zkEVM network experiences downtime, gameplay

continues uninterrupted. All gameplay state (level progress, star completions, hero experience) is tracked locally and synced to the blockchain when connectivity resumes. Blockchain-specific transactions (marketplace trades, token transfers, NFT operations) are queued locally and submitted automatically in order when the chain comes back online. This ensures that players never lose progress due to infrastructure issues.

21.3 Invisible Blockchain Philosophy

The invisible blockchain philosophy extends to every aspect of the game's design. There is no blockchain tutorial, no crypto jargon in the UI, and no requirement to understand gas, wallets, or smart contracts to enjoy the full gameplay experience. The UI uses standard gaming terminology ('Shop' instead of 'Marketplace,' 'Items' instead of 'NFTs,' 'Coins' instead of 'Tokens') for all core gameplay interactions. Blockchain-specific terminology is only used in the dedicated marketplace section, which players only see if they choose to explore trading features.

22. Multi-Account Prevention

Multi-account prevention is critical for the SAKART economy because each account can earn SAKART through star completions. Without prevention, a single person could create multiple accounts, complete the early (easier) levels on each account to farm SAKART, and then consolidate the tokens through marketplace trades. This would undermine the structural anti-inflation design and devalue the token for legitimate players.

The prevention system uses multiple layers. At the wallet level, Passport's identity system ties each account to a verified identity (email, phone, or social login), making it difficult to create large numbers of accounts. At the gameplay level, progression requirements (15/20 stars to unlock the next era) ensure that farming accounts must invest significant time before earning meaningful SAKART. At the transaction level, the marketplace monitors for patterns consistent with multi-account farming (multiple accounts sending tokens to a single address within a short timeframe) and flags suspicious activity for investigation.

The blacklist function in the SAKART token contract allows flagged accounts to be suspended from token transactions while preserving their gameplay progress. This ensures that legitimate players who happen to share IP addresses (family members, university networks) are not unfairly penalized. The threshold for blacklisting is intentionally conservative, requiring multiple indicators of farming behavior before action is taken.

23. Competitive Analysis

The competitive landscape for tower defense games includes established Web2 titles and a growing number of Web3 gaming projects. Sakartvelo Defenders differentiates itself through the unique combination of historical education, invisible blockchain integration, and a solo-developer-friendly PWA architecture. The following comparison highlights key differences between Sakartvelo Defenders and the most relevant competitors in the tower defense genre.

Feature	Sakartvelo Defenders	Kingdom Rush	Bloons TD 6	Arknights
Historical Theme	10 Georgian eras	Generic fantasy	None	Post-apocalyptic
Blockchain	Invisible (ERC-20/721/1155)	None	None	None
Platform	PWA (browser, any device)	Mobile only	Mobile + PC	Mobile + PC
Economy	Player-owned (SAKART)	Premium currency	Premium currency	Gacha system
Asset Ownership	True (NFT skins/scrolls)	None	None	None
Offline Play	Full campaign	Yes	Yes	Requires internet
Educational	Georgian history focus	None	None	Lore-based
Solo Dev Budget	$100/month feasible	Team required	Team required	Team required

23.1 Web2 Tower Defense Landscape

The Web2 tower defense market is dominated by Kingdom Rush, Bloons TD 6, and Arknights. These games represent the gold standard for tower defense gameplay and provide the benchmark for quality that Sakartvelo Defenders must meet. Kingdom Rush is known for its tight level design and tower upgrade system. Bloons TD 6 offers enormous content depth with hundreds of upgrades and game modes. Arknights combines tower defense with gacha collection mechanics and a rich narrative.

Sakartvelo Defenders borrows the best elements from each: Kingdom Rush's focused level design and upgrade clarity, Bloons TD 6's content volume and replayability, and Arknights' narrative engagement (adapted to historical rather than fictional storytelling). The key

differentiator is the historical education component, which provides a unique value proposition that none of these competitors offer. Additionally, the invisible blockchain integration provides true asset ownership without the friction that has prevented Web3 games from competing with Web2 quality.

23.2 Web3 Gaming Landscape

Web3 tower defense games are a nascent category with few established titles. Most Web3 games in this space have failed because they prioritize tokenomics over gameplay, resulting in products that are essentially financial instruments with game-like interfaces. Sakartvelo Defenders takes the opposite approach: the gameplay is designed to be competitive with Web2 titles, and blockchain is an invisible infrastructure layer that enhances the experience without defining it.

The invisible blockchain philosophy is the primary competitive advantage in the Web3 space. By eliminating wallet requirements, gas fees, and crypto jargon from the core experience, Sakartvelo Defenders can attract the much larger Web2 gaming audience while still providing Web3 benefits (true ownership, player-driven economy, transparent governance) to players who choose to engage with those features. This bridge between Web2 and Web3 audiences is the game's most significant market opportunity.

24. Art Pipeline

The art pipeline is designed around the constraints of solo development with AI-generated art. All visual assets are created using AI image generation tools, guided by a comprehensive Art Style Guide (see companion document) that ensures consistency across all 10 eras, 10 tower types, 200 maps, and hundreds of enemy and hero designs. The Art Style Guide provides era-specific color palettes, architectural reference keywords, character proportion standards, and prompt templates that enable reproducible, consistent output from AI tools.

24.1 Style Bible System

The Art Style Guide v2.0 (to be updated with the new 10-era structure) provides the authoritative reference for all visual assets. The guide defines a 2D isometric art style with cel-shaded lighting, a consistent character proportion system (10-unit body scale), era-specific color palettes (8 colors per era), and tower silhouette guidelines that ensure each tower type is visually distinct at small sizes. Every AI generation prompt includes era-appropriate keywords drawn from the style guide, ensuring that assets from different eras are visually cohesive within their period while being clearly distinct across periods.

24.2 AI Generation Workflow

The workflow begins with a prompt template that incorporates the style guide parameters for the target era and asset type. The prompt includes era-specific color palette references, architectural keywords (e.g., 'Svan medieval stone tower' for Era 4, 'Soviet brutalist concrete' for Era 8), and composition guidelines (isometric perspective, cel-shaded lighting, no text overlays). Generated images are reviewed against the style guide consistency checklist, which verifies color accuracy, proportion correctness, silhouette clarity, and era-appropriate architectural details.

Assets that pass the consistency check are composited into the game at the appropriate resolution. The isometric perspective and 2D art style make asset compositing straightforward, as all elements share the same perspective and lighting model. The style guide's standardization enables efficient batch generation: a single prompt template with era-specific parameter substitutions can produce an entire era's worth of tower skins, map backgrounds, and character art in a single generation session.

25. Tech Stack

The technology choices for Sakartvelo Defenders are driven by the constraint of solo development on a $100/month budget. Every technology in the stack is either free or within the budget envelope, and the architecture is designed to minimize ongoing costs while maximizing scalability. The combination of Godot for game development, PWA for deployment, and Immutable zkEVM for blockchain creates a complete, production-ready stack with zero server hosting costs.

Component	Technology	Rationale
Game Engine	Godot 4.x	Free, open-source, excellent 2D support, PWA export
Deployment	PWA (Progressive Web App)	Zero server costs, offline play, browser-based
Blockchain	Immutable zkEVM	Zero gas, EVM compatible, Passport wallet
Token Standard	ERC-20 (SAKART)	Well-established, wide tooling support
Skin Standard	ERC-1155	Fungible + non-fungible in one contract
Scroll Standard	ERC-721	Distinct NFT assets with metadata

Component	Technology	Rationale
AI Art	Stable Diffusion / DALL-E	Style guide-driven prompt architecture
Asset Storage	IPFS	Decentralized, persistent, censorship-resistant
Website	Vercel (static)	Free tier, fast CDN, automatic HTTPS
Budget	$100/month	Vercel free + Immutable free + domain only

25.1 Godot 4.x and PWA Architecture

Godot 4.x was selected as the game engine for several reasons: it is free and open-source, it has excellent 2D rendering capabilities, it supports PWA export natively, and it has a small but active community. The PWA deployment model means the game runs entirely in the browser with no server-side game logic. All gameplay computation happens on the player's device, eliminating the need for game servers. The only server interactions are blockchain transactions (handled by Immutable's infrastructure) and website hosting (handled by Vercel's free tier).

The PWA architecture also provides offline gameplay capabilities. The full campaign, all tower and hero data, and all map assets are cached locally through the PWA service worker. Players can continue playing the campaign even without internet access, with blockchain transactions queued for later submission. This offline capability is a significant competitive advantage over cloud-dependent games and aligns with the design philosophy of removing all friction from the player experience.

25.2 Blockchain Infrastructure

Immutable zkEVM provides the blockchain infrastructure with two critical advantages: zero gas fees for players and EVM compatibility. Zero gas is essential for the invisible blockchain philosophy, as any visible transaction cost would immediately break the illusion of a traditional game. EVM compatibility ensures that the smart contracts can use established standards (ERC-20, ERC-721, ERC-1155) and benefit from the extensive Ethereum tooling ecosystem.

Passport, Immutable's wallet abstraction layer, handles wallet creation and management. Players who have never used blockchain before can create a wallet through a simple email signup, while experienced Web3 users can connect their existing wallets. The Passport

system handles private key management, transaction signing, and network switching automatically, abstracting all blockchain complexity away from the player.

26. Design Principles

The following principles are the non-negotiable rules that govern all design decisions in Sakartvelo Defenders. When any design question arises, these principles provide the definitive answer. They are not suggestions or guidelines; they are hard constraints that cannot be overridden without a deliberate design review.

26.1 No-Nerf Rule

Once a tower, hero, scroll, or ability is released, its core functionality will never be nerfed (reduced in power). If something is too powerful, the response is to buff everything else to match, not to weaken the overperforming element. This principle protects player investment and builds trust. A player who spends time and resources mastering a specific strategy should never wake up to find that strategy has been intentionally weakened. The no-nerf rule is the most important design principle in the game.

26.2 Invisible Blockchain

No player should ever feel like they are interacting with a blockchain product unless they choose to engage with marketplace or token features. All UI language uses standard gaming terminology. No blockchain tutorial is required. No gas fees are visible to players. Wallet creation uses familiar social login methods. The blockchain exists as infrastructure, not as a feature that players must understand or engage with.

26.3 Historical Accuracy

The game is chronologically accurate and committed to truthful representation of Georgian history. Eras follow the correct historical sequence. Towers, enemies, and heroes are based on real historical references. The Stalin era is presented factually, with neither glorification nor erasure. When the historical record is ambiguous, the game acknowledges the uncertainty rather than presenting speculation as fact.

26.4 Solo Developer Feasibility

Every feature must be implementable by a solo developer on a $100/month budget. This constraint drives technology choices (PWA over dedicated servers, AI-generated art over hand-drawn assets, Immutable zkEVM over custom blockchain), content scope (10 eras with

20 levels each rather than open-ended content), and feature prioritization. Features that cannot be implemented within these constraints are deferred to post-launch community contributions or removed from the design entirely.

26.5 Accessibility

The game must be accessible to players with no prior experience with tower defense games, blockchain technology, or Georgian history. The tutorial system teaches core mechanics organically. Difficulty is calibrated to allow casual players to progress while providing meaningful challenges for dedicated players. Educational content is available but never forced. The blockchain is invisible to players who do not seek it out. Cultural references are explained in context rather than assumed as prior knowledge.

27. Roadmap

The roadmap is organized into six phases spanning approximately 19 months from project start to full launch. Each phase has clearly defined deliverables and budget requirements. The phased approach ensures that the game is playable at each stage (a minimum viable product after Phase 2, a blockchain-integrated product after Phase 3, and the full product after Phase 4), allowing for early playtesting and community feedback throughout development.

Phase	Timeline	Key Deliverables	Budget
Phase 1: Foundation	Months 1-3	Core gameplay loop, Era 0-1 (40 levels), basic UI, Godot PWA setup	$0 (free tools)
Phase 2: Expansion	Months 4-6	Era 2-4 (60 more levels), hero system, scroll system, synergy system	$0 (free tools)
Phase 3: Blockchain	Months 7-9	Immutable integration, SAKART contract, ERC-1155 skins, marketplace MVP	$0 (Immutable free tier)
Phase 4: Full Campaign	Months 10-14	Era 5-9 (100 more levels), all 10 tower types, PvP system, Survival Mode	$0 (free tools)

Phase	Timeline	Key Deliverables	Budget
Phase 5: Polish	Months 15-18	Art refinement, educational system, seasons, competitive features	$300 (domain + basic marketing)
Phase 6: Launch	Month 19	Full game launch, marketing push, community building, marketplace open	$100/month ongoing

27.1 Phase 1: Foundation (Months 1-3)

The foundation phase establishes the core game engine, the fundamental gameplay loop, and the first 40 levels of content (Era 0 and Era 1). Key technical deliverables include the Godot 4.x project setup, PWA export configuration, the tower placement system, enemy wave system, hero ability framework, and the basic UI/UX for level selection, gameplay HUD, and settings. Art deliverables include the Art Style Guide, era-specific color palettes for Era 0 and Era 1, and all tower and enemy visual assets for the first two eras.

By the end of Phase 1, the game should be a playable tower defense experience with 40 levels, 2 tower types (Archer Tower and Catapult/Trebuchet), 1 hero per era (Medea and King Pharnavaz), and the basic star-based progression system. This playable prototype will be used for initial playtesting and feedback collection, informing the design decisions for subsequent phases.

27.2 Phase 2: Expansion (Months 4-6)

The expansion phase adds 60 more levels (Era 2, 3, and 4), introduces the remaining initial tower types, implements the scroll system, and builds out the synergy system. This phase covers the most dramatic period of Georgian history, from the Arab invasions through the Golden Age to the Mongol Catastrophe, and requires the most diverse set of visual assets. The hero roster expands to include David IV, Queen Tamar, and the Bagrationi resistance heroes.

By the end of Phase 2, the game should have 100 playable levels across 5 eras, 6-7 tower types, 6-7 heroes, the full scroll system (both standard and mastery), and the synergy system with at least 10 active combinations. This represents a substantial playable experience that can be shared with a wider audience for testing.

27.3 Phase 3: Blockchain Integration (Months 7-9)

The blockchain phase integrates Immutable zkEVM into the existing game. Key deliverables include: SAKART ERC-20 token contract deployment, Passport wallet integration, ERC-1155 skin contract with marketplace, ERC-721 mastery scroll contract, and the marketplace UI with the confirmed tooltip system. This phase also implements the chain outage fallback system, ensuring that gameplay continues uninterrupted during any blockchain downtime.

By the end of Phase 3, the game should be a fully functional Web3-integrated tower defense game with 100 levels, real token economics, tradable cosmetic skins, and mastery scroll NFTs. The invisible blockchain philosophy must be validated: players with no blockchain experience should be able to play the full 100-level campaign without ever encountering crypto jargon or gas fees.

27.4 Phase 4: Full Campaign (Months 10-14)

The full campaign phase adds the remaining 100 levels (Era 5 through Era 9), completing the 200-level campaign. This phase also implements the final tower types (Gunpowder through Tech towers), the PvP system, and Survival Mode. The five eras in this phase cover the most diverse range of historical periods, from early modern warfare to the Soviet era to modern Georgia, requiring the broadest range of visual assets in the entire project.

The PvP system implementation follows the attack-then-defend format with RPS turn order determination, identical loadouts, no re-rolls, and AI-managed defenses. The competitive system must be balanced and fair from launch, with a league system organized by skill tiers. Survival Mode provides the endless challenge format with a global leaderboard.

27.5 Phase 5: Polish and Phase 6: Launch

The polish phase (months 15-18) focuses on art refinement, educational content completion, seasonal event system, and competitive feature tuning. The historical pop-up system is finalized with content for all 200 levels, the cultural footer is implemented, and the in-game encyclopedia is populated. The art pipeline is optimized for consistency, and any AI-generated assets that do not meet the style guide standards are regenerated.

The launch phase (month 19) is the public release of the complete game. Marketing efforts focus on tower defense communities, Web3 gaming forums, the Georgian diaspora community, and educational gaming advocates. The marketplace opens for full trading, the first season begins, and community management systems (Discord, social media) are fully operational. Post-launch development continues with seasonal content updates and community-driven feature requests.

28. Changelog (v2.0 to v3.0)

This section documents all changes made to the Game Design Bible between version 2.0 and version 3.0. The changes are driven by comprehensive Georgian history research that revealed significant historical inaccuracies in the previous era structure, user feedback on specific game systems, and design review feedback from Hermes. The v3.0 update is the most substantial revision to the Bible since the initial creation.

Section	Change	Reason
3: Eras	Expanded from 9 to 10 eras	First Democratic Republic kept separate (no merging)
3.2	Renamed era	Era 2: Age of Invasions replaces Golden Age of Svaneti
3.4	New era content	Era 3: Georgian Golden Age (David IV + Tamar, both heroes)
3.5	New era	Era 4: Mongol Catastrophe (1225-1500 AD) added
3.6	Renamed era	Era 5: Between Empires replaces Kakheti Wine Country
3.8	New era	Era 7: First Democratic Republic (1918-1921) separated
3.9	New era content	Era 8: Soviet Century (Stalin handled historically)
5: Enemies	Updated per-era enemy examples	Aligned with new chronological era structure
6: Heroes	Added David IV + Tamar as separate heroes	User confirmed: more heroes is better
6: Heroes	Stalin NOT a playable hero	Historical content only, factual presentation
7: Maps	Elevation mechanic confirmed	Towers on higher ground get +15-20% range
10: Scrolls	Removed SAKART Boost scroll	User confirmed: no SAKART boost
10: Scrolls	Mastery scroll stack system defined	3-stack max = +9% per type
11: Skins	Added marketplace tooltip requirement	Hermes feedback: item info transparency

Section	Change	Reason
12: PvP	No re-rolls, identical loadouts, AI defense	User-confirmed PvP design
16: Education	Added cultural footer requirement	Hermes feedback: cultural touch
16: Education	Historical pop-ups are free (no SAKART)	Hermes feedback: education should be free
18: Economy	Burn mechanisms expanded	Bind/unbind, seasons, cosmetics, potential reroll